

Операционные системы

Командная оболочка Midnight Commander

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2026-03-29

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

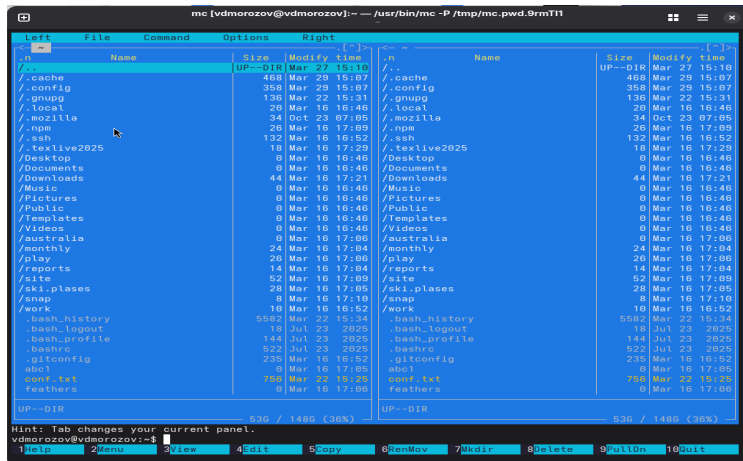


Рисунок 1: Запуск mc

Работа с Midnight Commander

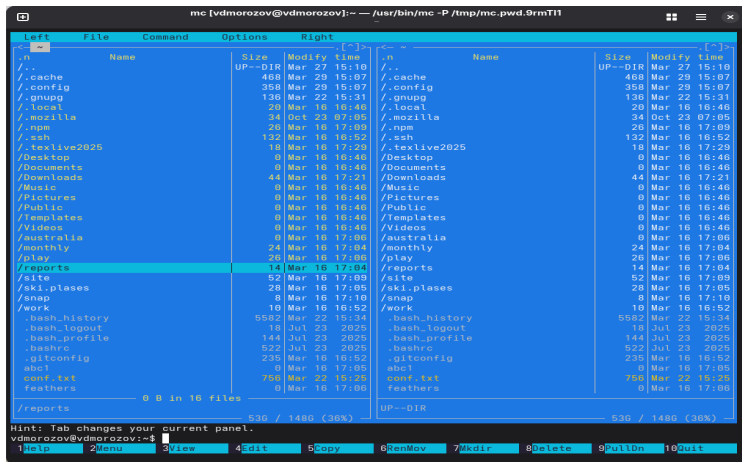


Рисунок 2: Выделение

Работа с Midnight Commander

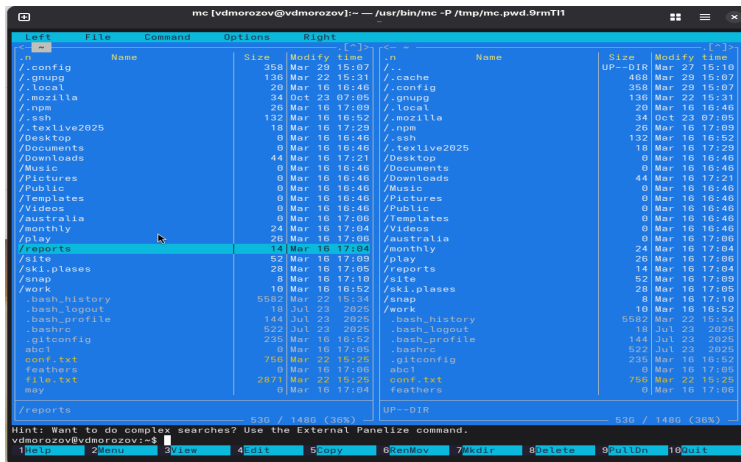


Рисунок 3: Отмена

Работа с Midnight Commander

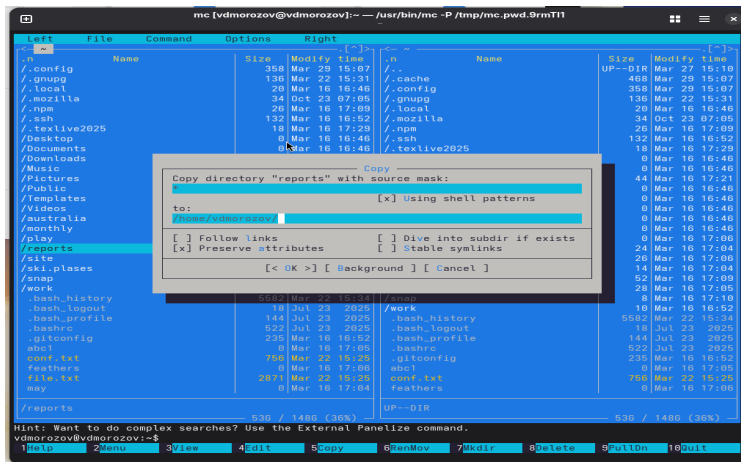


Рисунок 4: Копирование

Работа с Midnight Commander

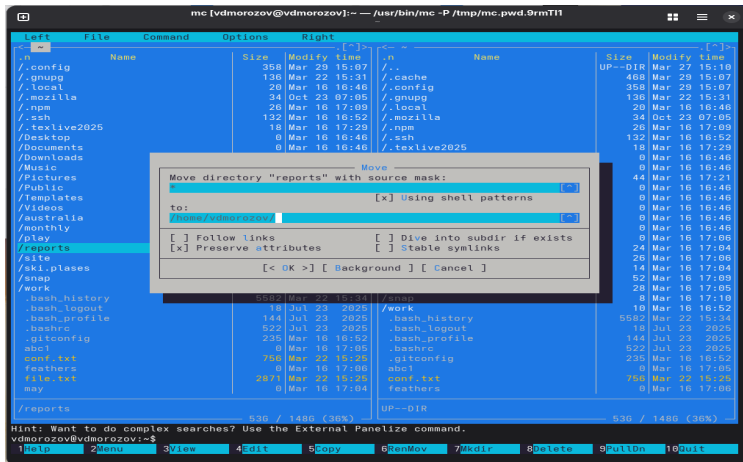


Рисунок 5: Перемещение

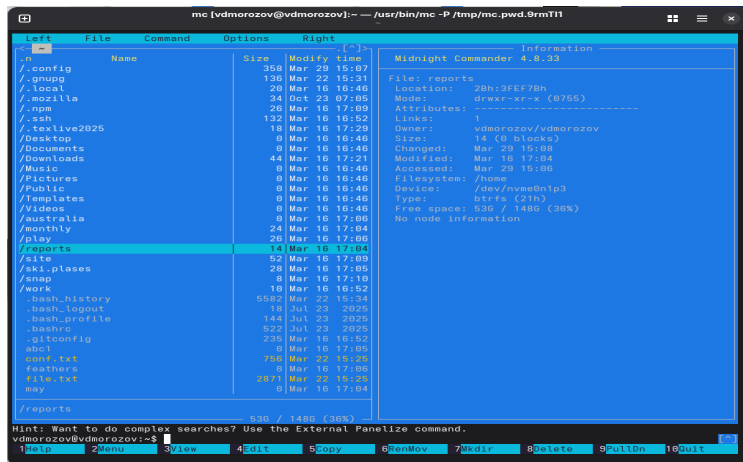


Рисунок 6: Информация

Работа с Midnight Commander

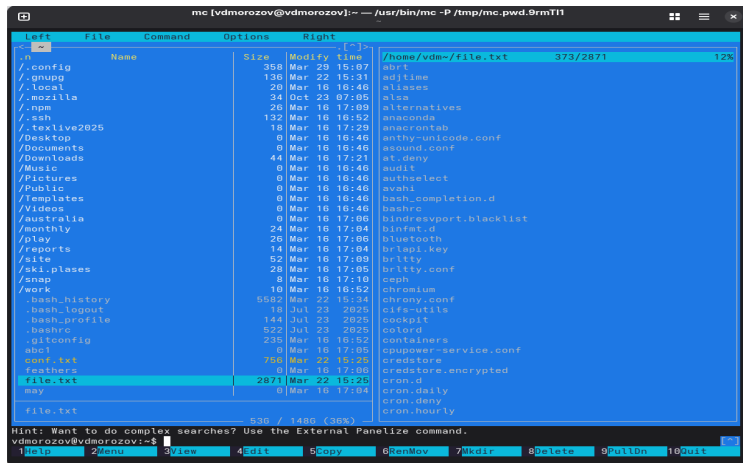


Рисунок 7: Быстрый просмотр

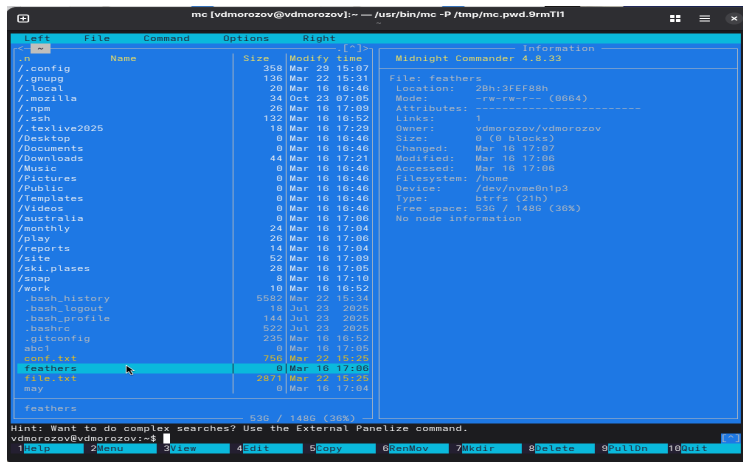


Рисунок 8: Информация

Работа с Midnight Commander

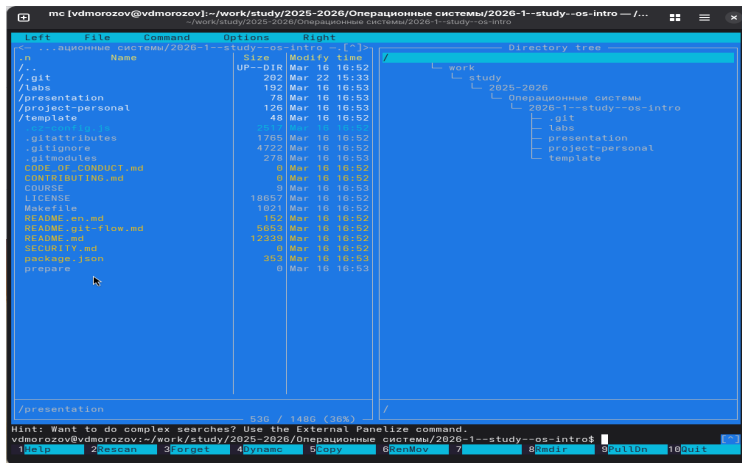
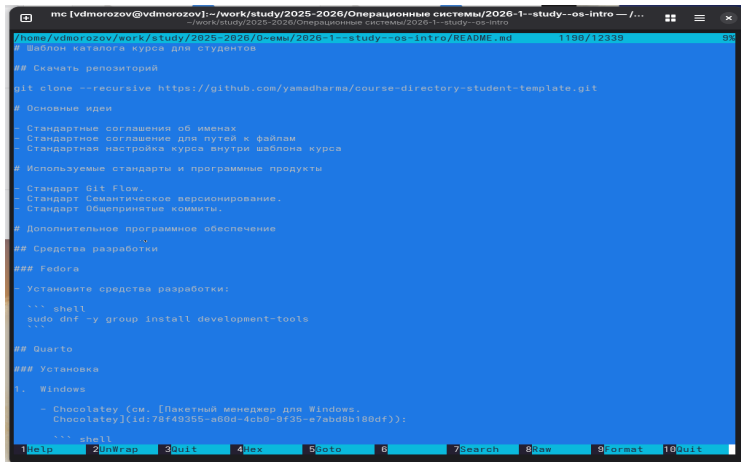


Рисунок 9: Дерево каталогов



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main window displays the contents of the `README.md` file, which includes instructions for cloning a repository and setting up development tools. The interface features a dark theme with a light blue background for the file content. At the bottom, there is a toolbar with numbered shortcuts for various actions like Help, UnWrap, Quit, Hex, Goto, Search, Raw, Format, and Quit.

```
mc [vdmorozov@vdmorozov]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro /...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
/home/vdmorozov/work/study/2025-2026/О-емы/2026-1--study--os-intro/README.md 1198/12339 9%
# Шаблон каталога курса для студентов

## Скачать репозиторий

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

# Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

# Используемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт семантическое versionирование.
- Стандарт общепринятые коммиты.

# Дополнительное программное обеспечение

## Средства разработки

### Fedora

- Установите средства разработки:

    ``` shell
 sudo dnf -y group install development-tools
    ```

## Quarto

### Установка

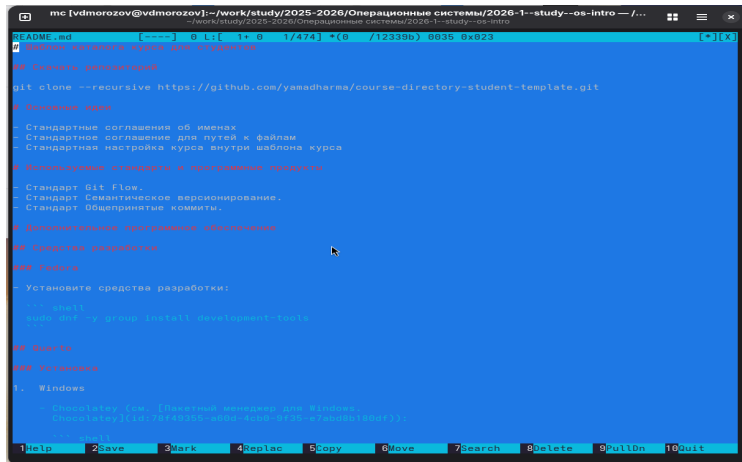
1. Windows

    - Chocolatey (см. [Пакетный менеджер для Windows.
      Chocolatey](id:78f49355-a60d-4cb0-9f35-e7abd8b180df)):

    ``` shell
```

Рисунок 10: Просмотр содержимого текстового файла

# Работа с Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main window displays the contents of a `README.md` file. The file content includes instructions for cloning a repository, setting up development tools, and installing Chocolatey on Windows. The interface features a dark theme with a blue background for the file content. At the bottom, there is a toolbar with numbered shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, and 10Quit.

```
mc [vdmorozov@vdmorozov]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro /...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
README.md [-----] 0 L: [1+ 0 1/474] *(0 /12339b) 8635 8x823 [*][X]
Шаблоны курсов для студентов

Сценарии развёртывания

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

Используемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт Семантическое versionирование.
- Стандарт Общепринятые коммиты.

Дополнительное программное обеспечение

Средства разработки

Fedora

- Установите средства разработки:

 ... shell
 sudo dnf -y group install development-tools

Quarto

VScode

1. Windows

- Chocolatey (см. [Пакетный менеджер для Windows.
 Chocolatey](id:78f48355-460d-4cb8-9f35-e7abd8b188df)).

 ... shell
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

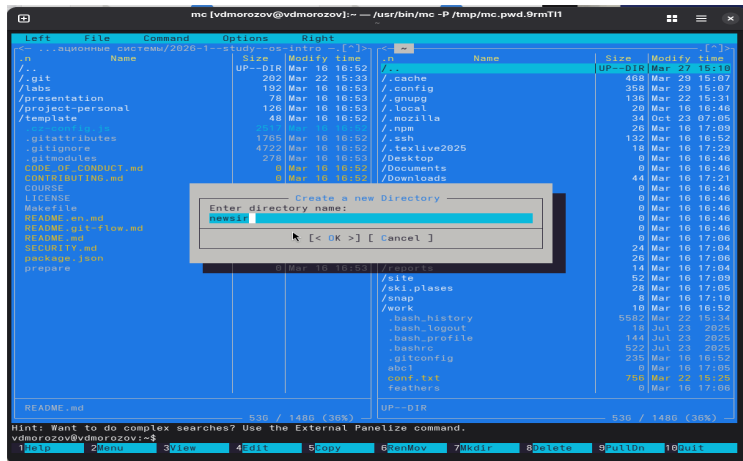
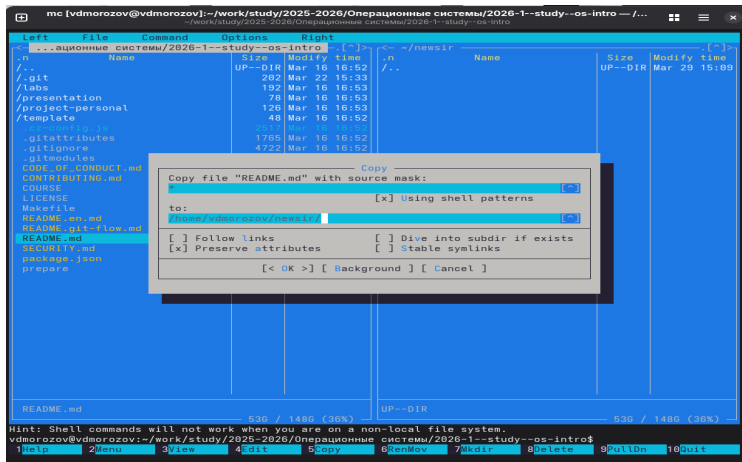


Рисунок 12: Создание каталога



**Рисунок 13:** Копирование в файлов в созданный каталог

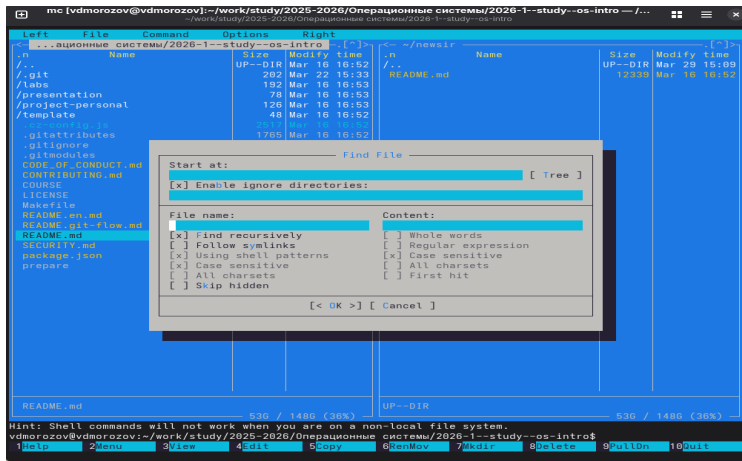


Рисунок 14: Поиск файлов

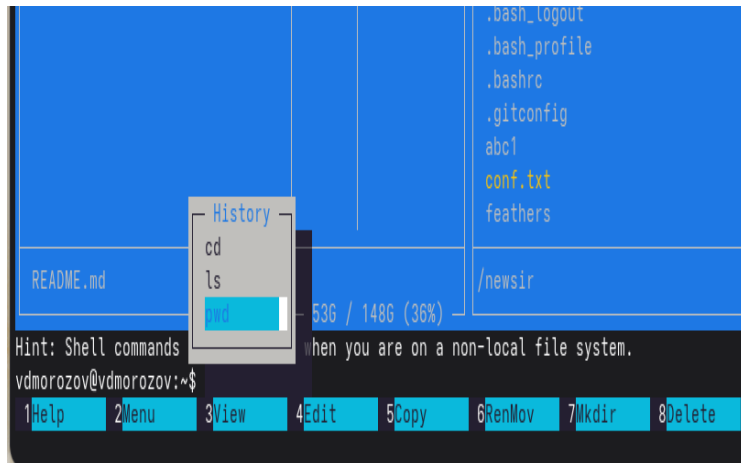


Рисунок 15: История команд

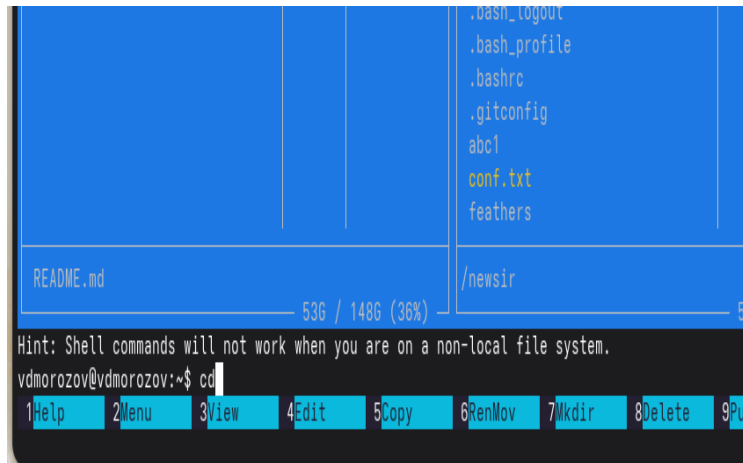
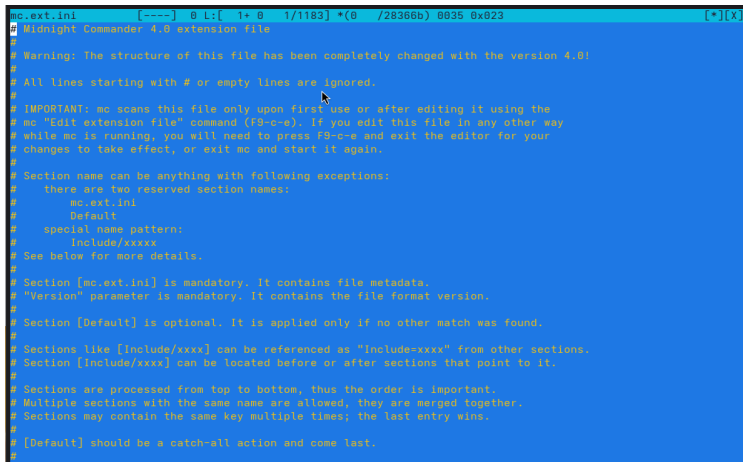


Рисунок 16: Переход в домашний каталог



```
mc.ext.ini [---] 0 L: 1+ 0 1/1183] *(0 /28366b) 0035 0x023 [*)(X)
Midnight Commander 4.0 extension file
#
Warning: The structure of this file has been completely changed with the version 4.0!
#
All lines starting with # or empty lines are ignored.
#
IMPORTANT: mc scans this file only upon first use or after editing it using the
mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
while mc is running, you will need to press F9-c-e and exit the editor for your
changes to take effect, or exit mc and start it again.
#
Section name can be anything with following exceptions:
there are two reserved section names:
mc.ext.ini
Default
special name pattern:
Include/xxxx
See below for more details.
#
Section [mc.ext.ini] is mandatory. It contains file metadata.
"Version" parameter is mandatory. It contains the file format version.
#
Section [Default] is optional. It is applied only if no other match was found.
#
Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
Section [include/xxxx] can be located before or after sections that point to it.
#
Sections are processed from top to bottom, thus the order is important.
Multiple sections with the same name are allowed, they are merged together.
Sections may contain the same key multiple times; the last entry wins.
#
[Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

```
menu [----] 0 L:[1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]
shell_patterns=0

#####
%% The % character
%f The current file (if non-local vfs, file will be copied locally and
%f will be full path to it)
%p The current file
%d The current working directory
%s "Selected files"; the tagged files if any, otherwise the current file
%t Tagged files
%u Tagged files (and they are untagged on return from expand_format)
%view Runs the commands and pipes standard output to the view command
If %view is immediately followed by '{', recognize keywords
ascii, hex, proff and unform
#
If the format letter is in uppercase, it refers to the other panel
#
With a number followed the % character you can turn quoting on (default)
and off. For example:
%f quote expanded macro
%if ditto
%0f don't quote expanded macro
#####

+ ! t t
0 Do something on the current file
 CMD=%{Enter command}
 $CMD %f

+ t t
0 Do something on the tagged files
 CMD=%{Enter command}
 for i in %t ; do
```

Рисунок 18: Просмотр файла меню

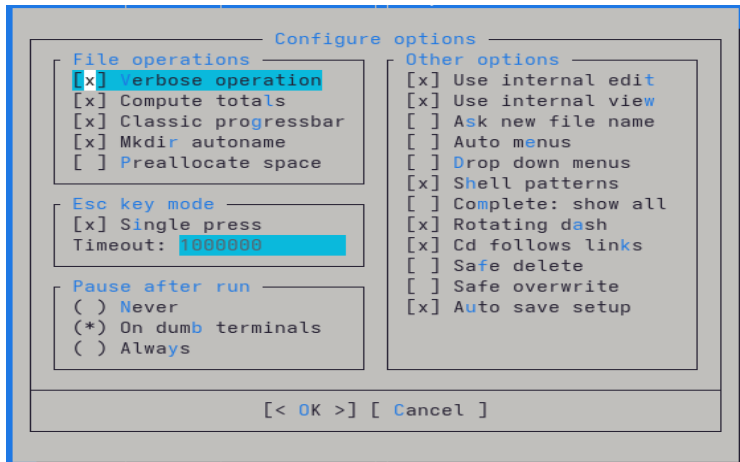


Рисунок 19: Конфигурация

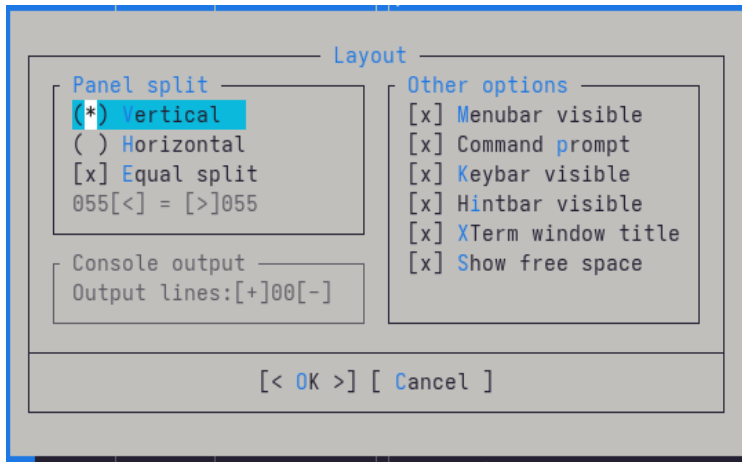


Рисунок 20: Внешний вид

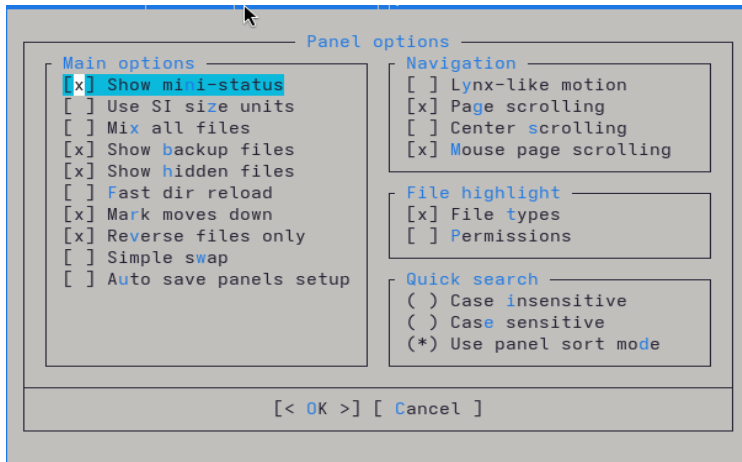


Рисунок 21: Настройки панелей

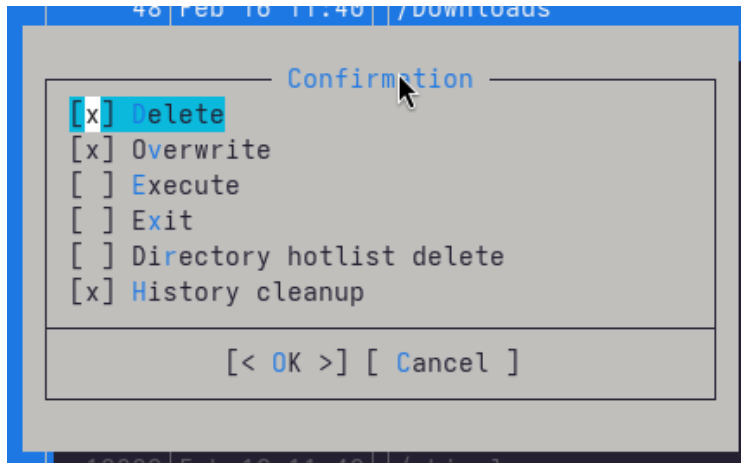


Рисунок 22: Подтверждение

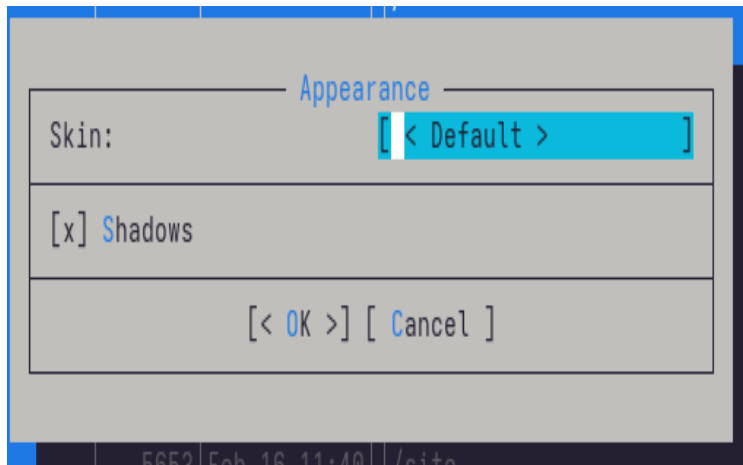


Рисунок 23: Оформление

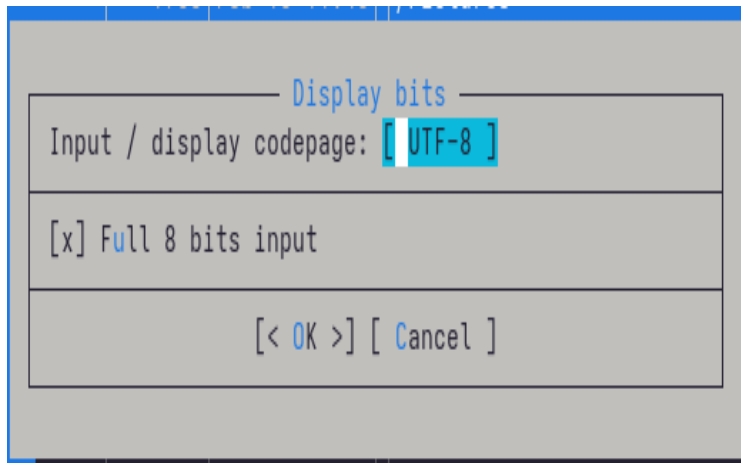


Рисунок 24: Кодировка символов

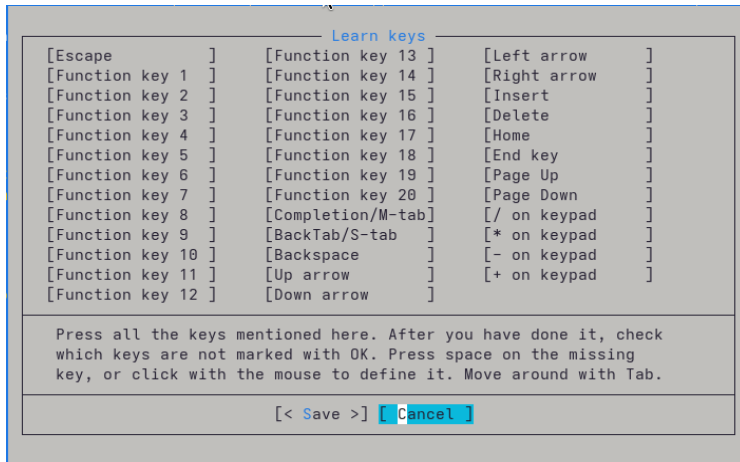
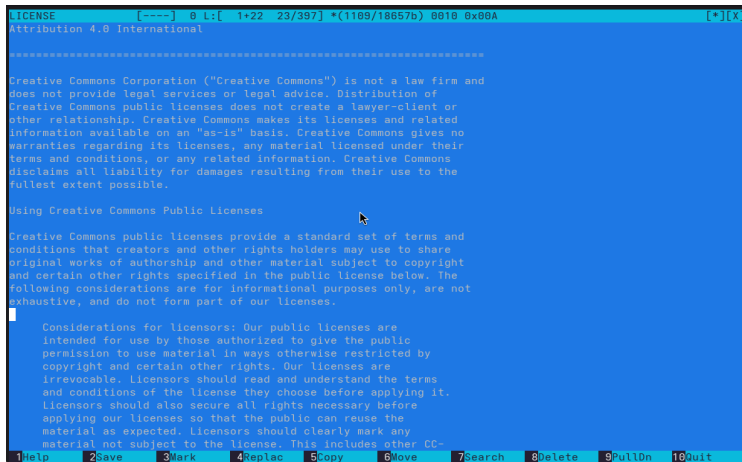


Рисунок 25: Распознавание клавиш



```
LICENSE [-----] 8 L: [1+22 23/397] *(1109/18657b) 0010 0x00A [*][X]
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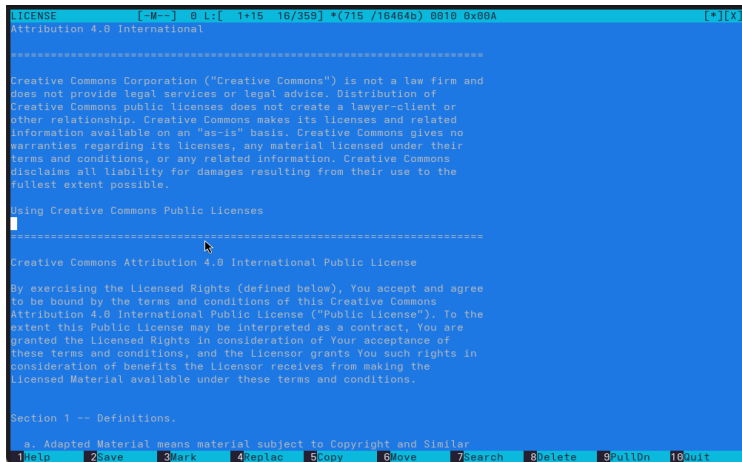
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 Licensors should also secure all rights necessary before
 applying our licenses so that the public can reuse the
 material as expected. Licensors should clearly mark any
 material not subject to the license. This includes other CC-
```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 26: Файл с текстом



```
LICENSE [-M--] 8 L:[1+15 16/359] *(715 /16464b) 0010 0x00A [*][x]
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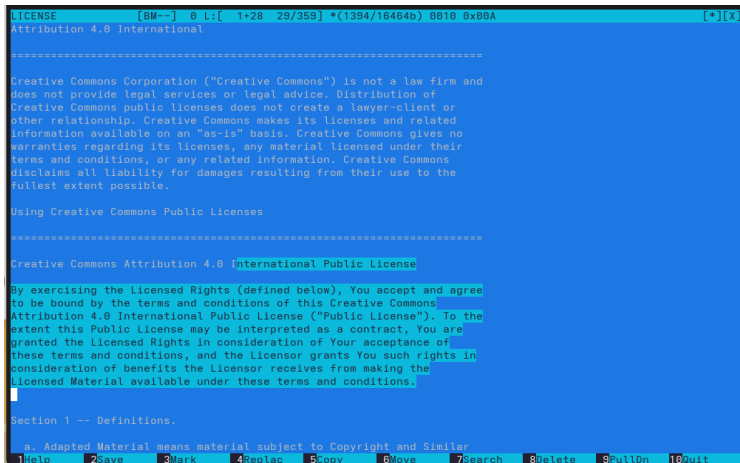
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 27: Файл с текстом



```
LICENSE [BM--] 0 L:[1+28 29/359] *(1394/16464b) 0010 0x00A [*][X]
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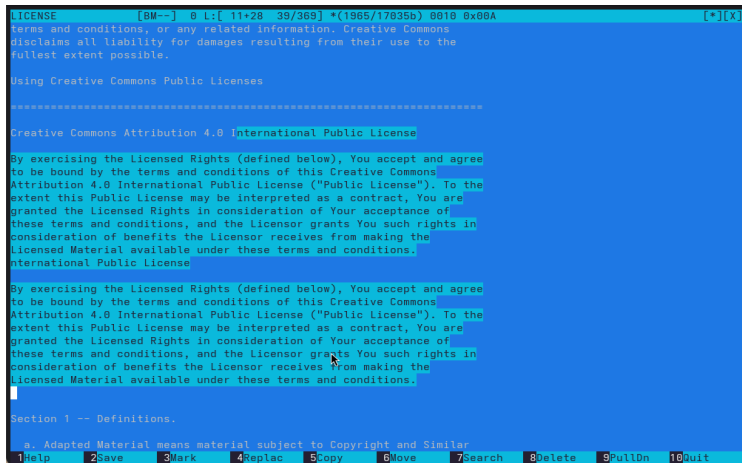
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullOn 10Quit
```

Рисунок 28: Копирование фрагмента



```
LICENSE [BM--] 0 L:[11+20 39/369] *(1965/17035b) 0010 0x00A [*][X]
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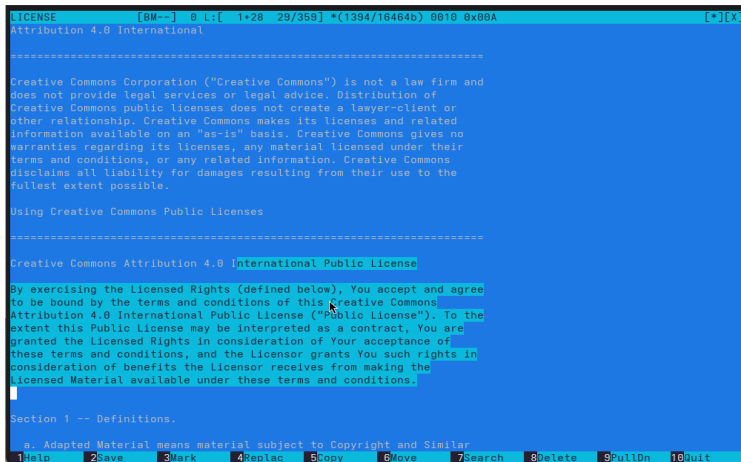
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 29: Сохранение

## Работа с редактором Midnight Commander



```
LICENSE [BM--] 0 L:[1+20 29/359] *(1394/16464b) 0010 0x00A [*][X]
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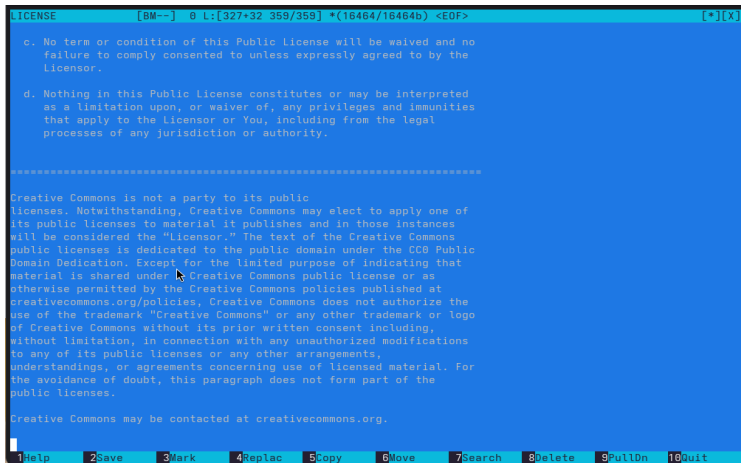
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 30: Отмена



```
LICENSE [BM--] 0 L:[327+32 359/359] *(16464/16464b) <EOF> [*][X]

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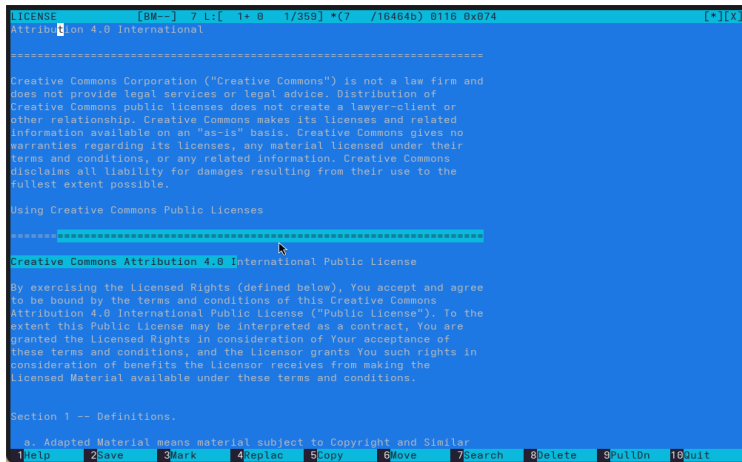
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 31: Переход в конец файла



```
LICENSE [BM--] 7 L:[1+ 0 1/359] *(7 /16464b) 0116 0x074 [*][x]
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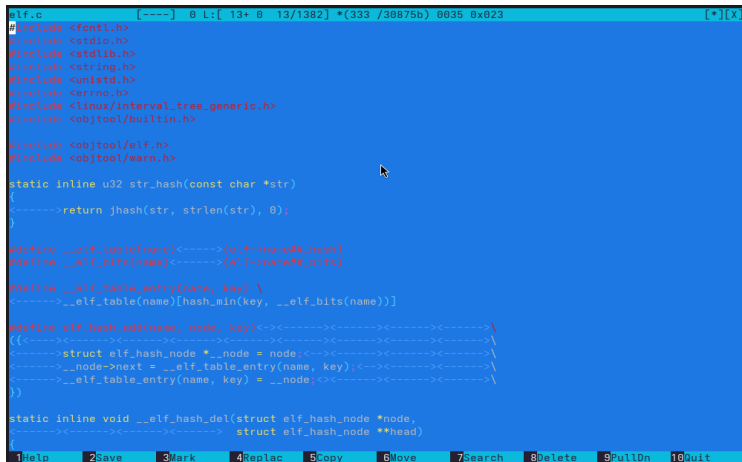
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Section 1 -- Definitions.

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```

Рисунок 32: Переход в начало файла



```
elf.c [----] 0 L: [13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
 ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
 ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->\
 ({----->----->----->----->----->\
 ----->struct elf_hash_node *__node = node;----->----->----->\
 ----->__node->next = __elf_table_entry(name, key);----->----->----->\
 ----->__elf_table_entry(name, key) = __node;----->----->----->\
 })

static inline void __elf_hash_del(struct elf_hash_node *node,
 ----->----->-----> struct elf_hash_node **head)
{
}
```

Рисунок 33: Файл с программой

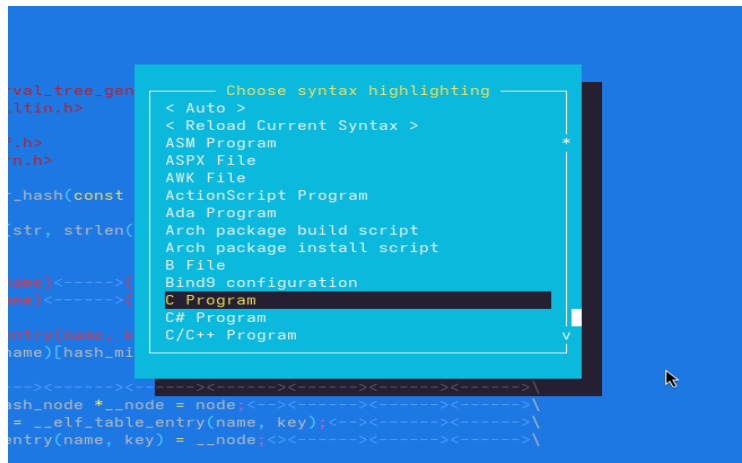


Рисунок 34: Цветовыделение синтаксиса

### 3. Выводы по проделанной работе

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В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.